

SVD Data Compression Using the Wine Dataset

Project II — Data Science and Engineering

Topic: Data Compression Using Singular Value Decomposition (SVD)

Dataset: Wine Dataset

1. Introduction

This project investigates the use of Singular Value Decomposition (SVD) for data compression using the Wine dataset. The main idea is to represent a numerical data matrix with a lower-rank approximation by retaining a selected number of singular components. The implementation is provided in a Jupyter Notebook using Python.

2. Objectives

The objectives are to load and inspect the Wine dataset, check for missing values, standardize the numerical features, apply SVD, examine the singular values, visualize the decomposition results, select a reduced number of components, and reconstruct the dataset using the selected components.

3. Dataset Description

The notebook loads the Wine dataset using `sklearn.datasets.load_wine()`. It contains 178 samples and 13 numerical features. The target values are also loaded, although the SVD compression is applied to the feature matrix. The feature names include alcohol, malic_acid, ash, alcalinity_of_ash, magnesium, total_phenols, flavanoids, nonflavanoid_phenols, proanthocyanins, color_intensity, hue, od280/od315_of_diluted_wines, and proline.

4. Data Inspection and Preprocessing

A pandas DataFrame is created from the feature matrix. The notebook checks for missing values and reports zero missing values for all 13 features. The feature matrix is then standardized with StandardScaler. Standardization is important because the Wine features have different numerical scales; for example, proline has much larger values than several other features.

5. Singular Value Decomposition

SVD is applied to the standardized matrix using NumPy. The decomposition is represented as $X = U\Sigma V^T$. The notebook obtains U with shape (178, 13), S with 13

singular values, and V^T with shape (13, 13). The diagonal matrix Σ is created from the singular values for reconstruction.

6. Singular Values

The singular values reported by the notebook are: 28.94203422, 21.08225141, 16.04371561, 12.78973645, 12.32374195, 10.68713954, 9.90368818, 7.87607330, 7.17081793, 6.68286180, 6.33958815, 5.48097635, and 4.28967045. The values decrease as the component index increases.

7. Visualization

Two visualizations are produced. The first plots singular values against r , allowing the distribution of the singular values to be observed. The second plots the cumulative sum of singular values against r , showing how the cumulative quantity increases as more components are included.

8. Selection of r

The notebook selects $r = 8$ for the compression experiment. The original decomposition has 13 components, while the reconstruction uses only the first eight singular components.

9. SVD Compression and Reconstruction

The compressed representation is constructed using U_r , S_r , and V_r^T , where only the first eight components are retained. The reconstructed matrix is calculated as $X_r = U_r \Sigma_r V_r^T$. The original standardized matrix has shape (178, 13), and the reconstructed matrix also has shape (178, 13). The reduction is therefore in the number of SVD components used to represent the matrix, not in the displayed dimensions of the reconstructed matrix.

10. Results and Discussion

The experiment demonstrates that the Wine feature matrix can be reconstructed from a reduced set of SVD components. The first eight singular values are retained while the remaining five components are excluded from the reconstruction. The notebook compares the first five rows of the standardized original data and the reconstructed data so that the effect of the lower-rank approximation can be observed.

11. Conclusion

This project demonstrates the application of SVD as a numerical data-compression technique. The Wine dataset was standardized, decomposed into singular components, and reconstructed using $r = 8$ instead of all 13 components. The resulting reconstruction keeps the original matrix dimensions while using a lower-rank representation. This illustrates the central principle of SVD-based compression: important structure can be represented using fewer components, with some information omitted as the rank is reduced.

12. Tools and Libraries

Python, Jupyter Notebook, NumPy, Pandas, Matplotlib, and Scikit-learn are used in the project.

13. Key Results Table

Item	Before	After
Number of samples	178	178
Number of features	13	13
Number of SVD components	13	8
Matrix used for reconstruction	Standardized X	X_r using first 8 components

Source basis: This report was prepared from the supplied Jupyter Notebook, “Wine dataset Compression.ipynb,” including its code, outputs, descriptions, and selected results.